

Mobile-Based Convolutional Neural Network Approach for Maize Leaf Disease Detection

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Abstract—Although maize is a major food crop in the world, foliar diseases like blight, common rust, and grey leaf spot pose a serious threat to its output. In order to minimise crop losses and guarantee food security, early detection of these diseases is crucial. Manual inspection techniques are not appropriate for widespread use since they are frequently expensive, time-consuming, and necessitate specialised skills. This study uses deep convolutional neural networks (CNNs) to develop a real-time, mobile-based system for detecting and classifying maize leaf disease. Images of maize leaves from four classes—Blight, Common Rust, Grey Leaf Spot, and Healthy—were used from the CV-Corn-Disease-Detection dataset (GitHub). Techniques for data augmentation and preprocessing were used to improve image quality and balance the dataset. VGG16, Mobilnet, and ResNet18 were among the CNN architectures that were trained and contrasted. Accuracy, precision, recall, F1-score, confusion matrix were used to assess the model's performance. To give farmers and other agricultural stakeholders a useful, approachable tool for early crop protection and real-time disease diagnostics, the top-performing model was included into a mobile application.

Index Terms—Maize leaf disease detection, Convolutional Neural Networks, CV-Corn-Disease-Detection dataset, Mobile application, Performance evaluation

I. INTRODUCTION

One of the most extensively grown cereal crops in the world, maize (*Zea mays* L.) is important for industrial uses, animal feed, and food security. Foliar diseases like blight, common rust, and grey leaf spot, however, commonly threaten its output and can result in significant productivity losses if not identified early. Manual observation is the foundation of traditional diagnosis techniques, which are labor-intensive, slow, and reliant on the availability of experts.

Automated methods for plant disease identification have been made possible by developments in artificial intelligence (AI), especially in the area of deep learning techniques like Convolutional Neural Networks (CNNs). Although the majority of previous systems function offline and are not easily available to farmers in the field, CNN-based models have shown impressive performance in crop disease classification tasks [1].

The current study fills this gap by creating a mobile-based system for detecting maize leaf disease that can offer an easy

to use, real-time diagnosis. The study utilized the CV-Corn-Disease-Detection dataset [2], which included photos of maize leaves categorised into four groups: healthy, common rust, blight, and grey leaf spot. Techniques for preprocessing and augmentation were used to improve the quality of the data and lessen class imbalance. Accuracy, precision, recall, F1-score, confusion matrix were used to evaluate the performance of many CNN architectures, including VGG16, MobileNet, and ResNet18.

This research supports farmers in sustainable crop production and food security by providing a practical solution for the early diagnosis and management of maize leaf diseases by integrating the best performing model into a smartphone application.

II. RELATED WORK

There have been several attempts to identify and categories illnesses of maize. A model for predicting maize leaf disease based on a picture of an afflicted maize leaf [3] uses

A deep learning method for identifying and categorising illnesses of maize leaves. Three cutting-edge CNN models (AlexNet, ResNet50, and SqueezeNet) were evaluated by Erik et al. [1] for their ability to categories maize illnesses into four groups: healthy, common rust, leaf bight, and grey leaf spot. The researcher evaluated the model performance using Bayesian hyper parameter optimization, fine-tuning techniques, data augmentation, and fivefold cross validation procedures, and all models obtained 97% accuracy. In an attempt to categories common maize leaf spot, grey spot, and rust diseases, a YOLOv5n-based model was presented by [4]. The core of YOLOv5n incorporates coordinate attention to increase model accuracy, which was attained at 94.5%.

However, a mobile device-applicable real-time maize disease detection and classification model is needed [5]. [5] Harry et al. uses adaptive thresholding to measure leaf diseases in maize. The researchers determine the degree of picture leaf illness with 99% accuracy by training deep

learning models, comparing their output, creating adaptive thresholding strategies, and using regions of interest. Using EfficientNet, Fathimathul et al. [6] are able to classify common rust, blight, and the northern leaf grey spot with high accuracy. The researcher obtained 98.5% accuracy by using preprocessing techniques such as image scaling, data augmentation, and classification. But in order to improve the model's accuracy, it is necessary to increase the size of the data, add more plant species to make the model more flexible, and create a mobile application that is easier to use and more accessible so that farmers can use it in the field to detect and identify disease in real time [6].

With an accuracy of 98.5%, the Mobile-DANet model put forth by [7] can successfully identify maize diseases. It was noted that in order to identify diseases in real time, the model must be deployed on portable devices. [8] used the ResNet-18, VGG16, and EfficientNet models to classify maize illness, with 97.1% accuracy.

To lessen class imbalance, a weighted cross-entropy loss approach is employed [8]. [9] examined seven convolutional neural network models to categorise maize leaf illnesses into common rust, northern leaf blight, cercospora, and healthy: AlexNet, VGG16, VGG19, GoogleNet, InceptionV3, ResNet50, and ResNet101. The classification performance was also compared using K-support vector machine and K-nearest neighbour machine learning classification techniques; AlexNet and Support vector machine produced the best results [9]. A pre-trained convolutional neural network-based Android recognition tool for identifying maize diseases [10] has a 93.4% classification accuracy for common rust, grey leaf spot, northern leaf spot, and healthy. Manta-Ray Foraging Optimisation with convolutional deep learning [11] produced an intelligent maize leaf disease detection system that included MRFO for hyperparameter tuning, DenseNet for feature extraction, and a median filter for image preprocessing. A convolutional neural network approach was used to identify Cercospora leaf spot, Common rust, and Northern leaf blight in grey maize [12]. The model, which was created using 4000 publically accessible photos and run in the MATLAB environment, has an accuracy rate of 96.53% [12]. The majority of researchers are focused on creating convolutional neural network models for maize disease detection. However, farmers need a real-time maize disease detection model that is simple to use, allows them to monitor, and provides them with instant feedback.

III. METHODOLOGY

Data collection, data preprocessing, suggested model design and selection, and mobile app development are all included in the suggested technique. The accompanying figure (Fig. 1) shows our full technique.

INPUT IMAGE

The GitHub repository provided a total of 4188 colour (RGB) photos of maize leaves [2]. Four categories make up the dataset: Common Rust (1306 photos), Gray Leaf Spot (574 images), Blight (1146 images), and Healthy (1162 images) in (Fig 2.A). According to the bar chart in (Fig 2.B), which displays the dataset's class distribution, the Common Rust class has the most samples, followed by Healthy and Blight, while Gray Leaf Spot has comparably less photos. The proportionate distribution of photos is shown in the accompanying pie chart, where Common Rust accounts for the highest percentage (31.19%), followed by Healthy (27.74%) and Blight (27.37%), while Gray Leaf Spot makes up just 13.70%. This distribution shows that the Gray Leaf Spot class is still under-represented even if the dataset is fairly balanced across the three classes

PREPROCESSING

Several preprocessing techniques were used to improve the dataset's quality and classification performance. The dataset was separated into subsets for testing, validation, and training. To ensure consistency, all photos were resized to 224×224 pixels. In order to promote faster convergence during training, image pixel values were normalised to the interval $[0, 1]$. Data augmentation techniques, such as random rotations up to 25° , width and height shifts, shear, zoom transformations, and horizontal flipping, were used to improve diversity in the training set and decrease overfitting. To maintain structural integrity, nearest-neighbor filling was used. In order to remove noise from extraneous pixels and highlight pertinent leaf sections, background segmentation was also carried out. In order to extract leaf areas, photos had to be converted to HSV colour space, a green-color mask applied, and the mask refined using morphological processes. Shuffling was turned off to maintain ordering for the validation and testing subsets, and only rescaling was used to guarantee consistency throughout evaluation. For the categorisation of maize leaf diseases, these preprocessing procedures guaranteed robustness, enhanced generalisation, and offered a solid basis.

DEVELOPING DEEP CNN ARCHITECTURES FOR THE DETECTION AND CLASSIFICATION OF MAIZE LEAF DISEASE

Three deep CNN architectures—VGG16, ResNet18, and MobileNetV2—were used in this study to identify and classify maize leaf disease. These architectures were chosen because they have a track record of successfully classifying images and are appropriate for datasets pertaining to plant diseases. The convolutional base was frozen and extra layers were added, such as a global average pooling layer, fully connected dense layers with ReLU activation, and dropout regularisation (0.5 and 0.4) to lessen overfitting prior to the final softmax classification layer. VGG16 was utilised as a transfer learning backbone with ImageNet weights.

After initialising ResNet18 with pretrained weights, a new linear layer matching to the number of classes was added in place of the last fully connected layer. The Adam optimiser was used for training with a learning rate of 0.0001, cross-entropy loss, and a StepLR scheduler (step size 7, gamma 0.1) to dynamically modify the learning rate.

After being rescaled using ImageNet weights and optimised for the maize dataset, MobileNetV2 was chosen due to its lightweight and effective architecture. Early stopping was utilised to avoid overfitting, categorical cross-entropy was utilised as the loss function, and ReLU activation functions were applied in the hidden layers of all models. Depending on the design, convergence was frequently seen within 10–20 epochs of the models being trained for up to 40 epochs. This configuration guaranteed a fair trade-off between classification accuracy, computing efficiency, and robustness.

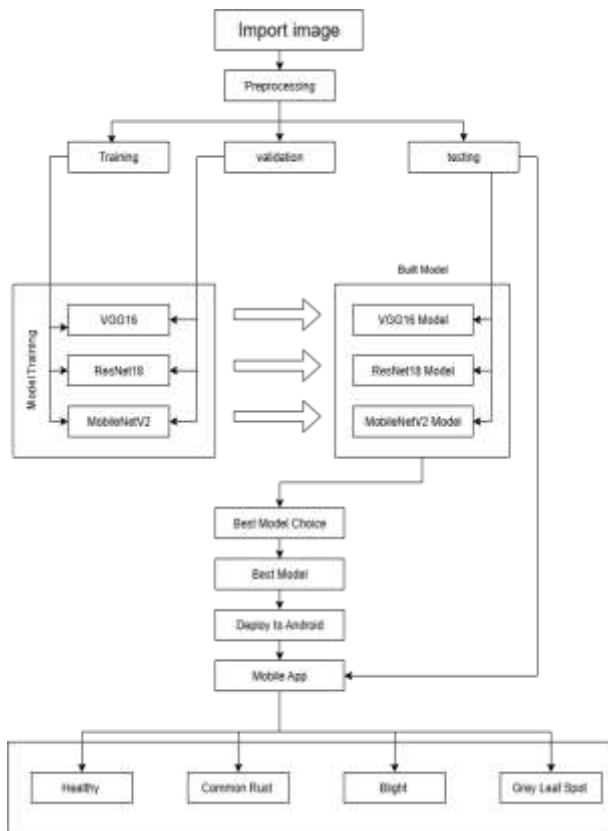


Fig. 1 Proposed model architecture

The maize leaf dataset was initially divided into subgroups for testing (10%), validation (20%), and training (70%). This ensured a systematic approach to model construction and assessment. Images were preprocessed and scaled to 224×224 pixels for homogeneity across all three CNN designs.

VGG16

The convolutional foundation was frozen and further layers, such as a global average pooling layer, dense layers with ReLU activation, and dropout regularisation (0.5 and 0.4), were added using the VGG16 architecture with ImageNet pretrained weights. The four types of maize diseases were represented by

the final softmax classifier. While validation and testing photos were just rescaled, training data were enhanced with random rotations (25°), width and height changes (0.2), shear, zoom, and horizontal flipping. The Adam optimiser (learning rate = 0.0001) and categorical cross-entropy loss were used to assemble the model. Class weights were calculated and used during training in order to solve dataset imbalance. To guarantee stable convergence, the model was trained for ten epochs with early stopping.

MOBILENETV2

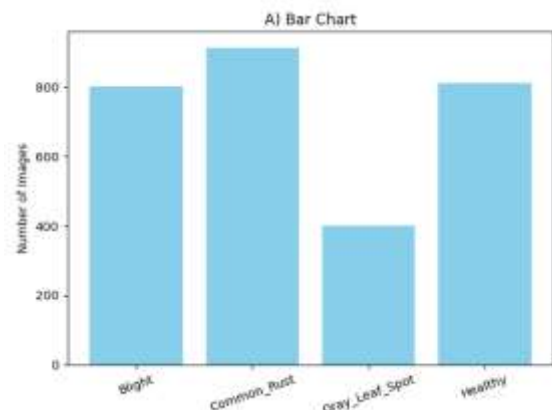
This lightweight substitute, which is optimised for efficiency and mobile deployment, was embraced. The convolutional base was frozen and pretrained ImageNet weights were applied. For classification, a softmax classifier, dense layers with ReLU activation, dropout regularisation, and a global average pooling layer were included. As in Model 1, the training, validation, and test sets underwent preprocessing; only the training set underwent augmentation. The model was trained for up to 20 epochs with validation monitoring using Adam (learning rate = 0.0001) and categorical cross-entropy.

RESNET18

ResNet18 was used to mitigate vanishing gradient issues that are typical in deeper networks by utilising residual learning. Pretrained weights were used to initialise the model, and a new linear layer representing the four classes was added in lieu of the last completely connected layer. Similar to the other models, preprocessing and augmentation were applied to images from the training, validation, and testing sets. Cross-entropy loss, the Adam optimiser (learning rate = 0.0001), and a StepLR scheduler (step size = 7, gamma = 0.1) for adaptive learning rate modification were used during training. After up to 40 epochs of training, the network frequently showed convergence in the range of 10 to 15 epochs.

No.	Type	Number(size)
1	Blight	1146
2	Common rust	1306
3	Gray leaf spot	574
4	Healthy	1162
Total		4188

Table 1 Detail of maize disease dataset utilized in this study



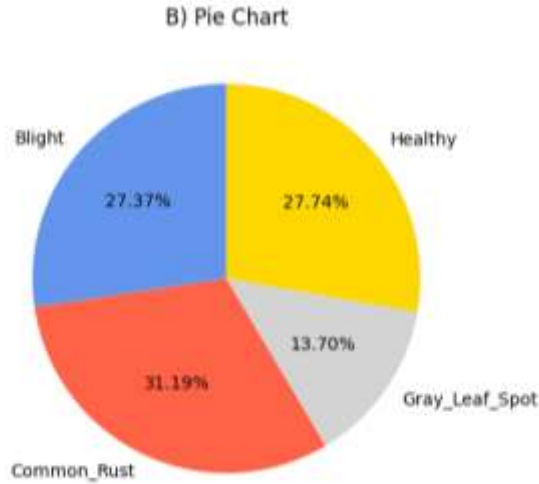


Fig.2 A) Number of images in each class using bar chart and B) Proportion of images in each class using pie chart

IV. EXPERIMENT

In this stage, we used photos of maize leaf disease to train the VGG16, MobileNetV2 and ResNet18 tree pre-trained architectures. After that, we construct the model and use test photos to assess its performance. Ultimately, we used the model that performs better than the other to create a mobile application.

On a local machine equipped with an Intel® Core™ i7-6600U CPU, 8 GB of RAM, and an integrated Intel® HD Graphics 520, the models were developed in Python. Ten percent was used for testing, twenty percent for validation, and seventy percent for training. While validation and testing sets were just rescaled, training images were preprocessed with normalisation (pixel values scaled to [0,1]) and enhanced with random rotations, shifts, zoom, shear, and horizontal flips. The Adam optimiser (learning rate = 0.0001) and categorical cross-entropy loss were used to construct each model. Each model was trained for up to 10–40 epochs, depending on convergence, with validation monitoring and the best weights kept for testing. Early termination was used to avoid overfitting.

EVALUATION METRICS

Every model is evaluated independently, and we have utilised common performance measure metrics including precision, recall, F1-score, accuracy, and confusion matrix to assess the suggested model. These assessment criteria are applicable to multi-class classification issues and are well suited for classification tasks.

The following mathematical formula and explanation of precision, recall, f1-score, and accuracy are based on research by [43, 45, 46].

A classification model's recall quantifies its capacity to accurately locate every pertinent occurrence of a given class in a dataset. Recall in mathematics is provided by Equation (1).

$$\text{Recall} = \frac{\text{TruePositives}}{\text{TruePositives} + \text{FalseNegatives}} \quad (1)$$

Precision measures how well a classification model predicts the favourable outcomes. The ratio of true positive predictions to the total of true positive and false positive predictions is used to compute precision. Equation (2), which provides mathematical precision, is provided.

$$\text{Precision} = \frac{\text{TruePositives}}{\text{TruePositives} + \text{FalsePositives}} \quad (2)$$

Equation (3), which provides the F1-score, can be understood as a weighted harmonic mean of the precision and recall.

$$\text{F1-score} = \frac{2 * (\text{Precision} * \text{recall})}{\text{Precision} * \text{recall}} \quad (3)$$

The proportion of accurate predictions to total predictions is known as accuracy. Provided mathematically in Equation (4).

$$\text{Accuracy} = \frac{\text{TruePositives} + \text{TrueNegatives}}{\text{TruePositives} + \text{FalsePositives} + \text{TrueNegatives} + \text{FalseNegatives}} \quad (4)$$

Furthermore, the confusion matrix is used to evaluate the performance of the suggested model. The correctly classified TP values, FP values that belong to the incorrect class, false negative (FN) values that belong to the incorrect class, and correctly classified TN values in the opposite class are all comprehensively represented by the confusion matrix [43]. In order to characterise how well a classification model performs on the set of test data for which the true values are unknown, we have employed a confusion matrix as a performance metric.

EXPERIMENTAL RESULT

In this stage, we assess the ResNet18 model's performance on the classification job for maize leaf disease. The training and validation accuracy curves are continuously high, as seen in the figures; after ten epochs, the training accuracy approaches 100% and the validation accuracy stabilises between 94 and 95%. This suggests that the model learns efficiently without experiencing extreme overfitting and while retaining good generalisation.

Strong optimisation is demonstrated by the loss curves, which also show proper convergence. The training loss rapidly decreases while the validation loss stabilises between 0.18 and 0.20 as illustrated in Fig.3.

With an overall accuracy of 95%, the classification report further highlights ResNet18's strong performance. With

precision, recall, and F1-score all equal to 1.00, the Healthy class performed the best, showing flawless classification with no misclassifications. Reliable detection was confirmed by the model's good performance on the Common Rust class, which included precision of 0.98, recall of 0.96, and F1-score of 0.97. While the Gray Leaf Spot class was relatively more difficult, achieving 0.89 accuracy, 0.84 recall, and 0.87 F1-score, the Blight class demonstrated balanced performance with 0.90 precision, 0.93 recall, and 0.91 F1-score is given in the Table 2.

ResNet18 performs exceptionally well overall, with high accuracy and good classification ability across the majority of classes; nevertheless, Gray Leaf Spot case differentiation still requires work.

C) ROC curve of ResNet18

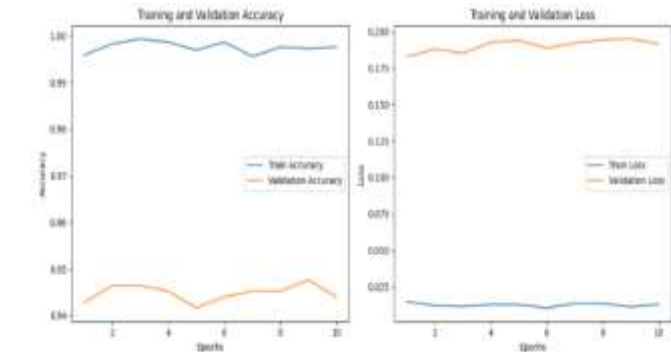
Fig. 3 ResNet18 performance on maize disease classification

Classification Report:

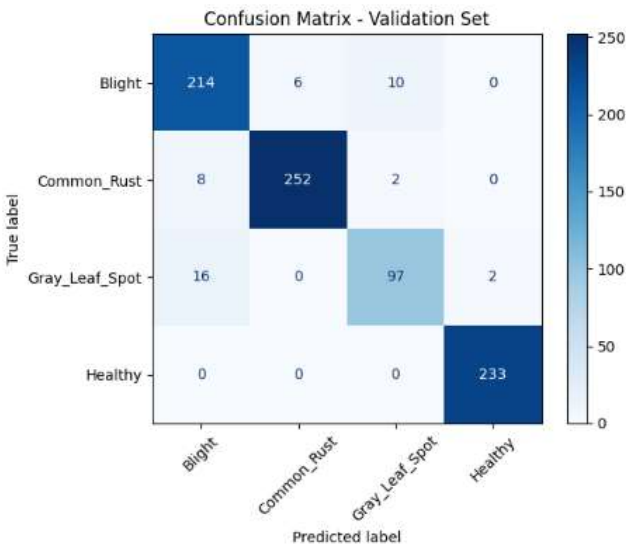
precision	recall	f1-score	support	
Blight	0.90	0.93	0.91	230
Common_Rust	0.98	0.96	0.97	262
Gray_Leaf_Spot	0.89	0.84	0.87	115
Healthy	0.99	1.00	1.00	233
accuracy			0.95	840
macro avg	0.94	0.93	0.94	840
weighted avg	0.95	0.95	0.95	840

Table 2. Classification report of ResNet18

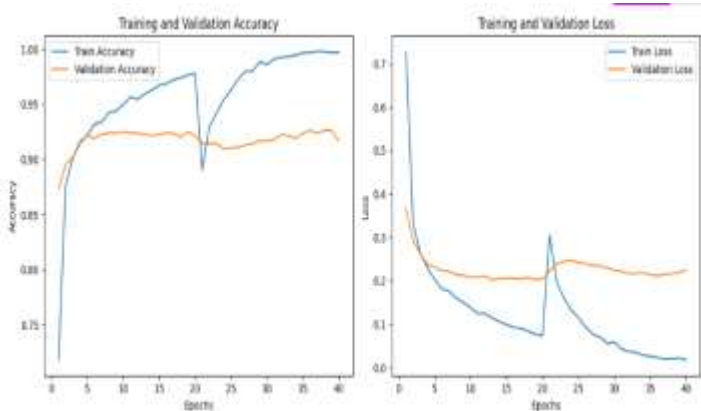
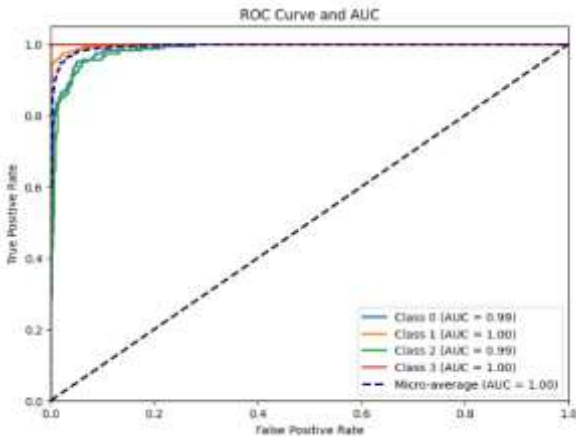
MobileNetV2's performance on the maize leaf disease classification test illustrated in Fig.4. With both measures stabilising at 92% after 20 epochs, the training and validation accuracy curves show that the model converges smoothly, indicating efficient learning without appreciable overfitting. The model is clearly optimising, as evidenced by the loss plots, which show a consistent drop in both training and validation loss. The categorisation report also demonstrates MobileNetV2's resilience, with an overall accuracy of 91.6%. With a precision, recall, and F1-score of 1.00, the Healthy class performed the best, proving that the model is capable of correctly classifying healthy maize leaves given result in Table 3. Additionally, Common_Rust was categorised with great reliability, demonstrating 98% precision and 95% recall.



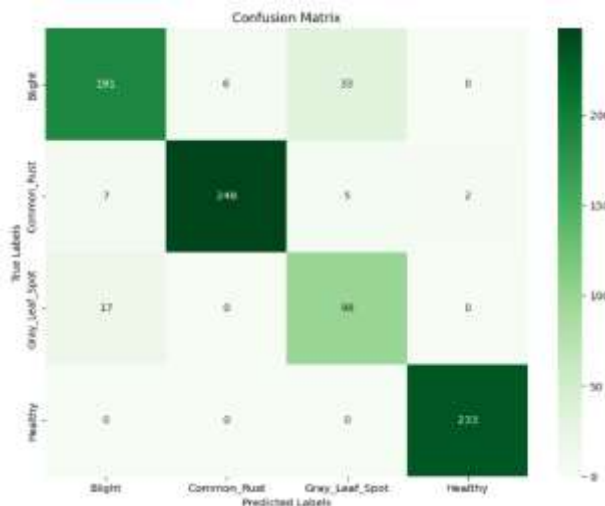
A) Accuracy and Loss Plot of ResNet18.



B) Confusion Matrix of Res



A) Accuracy and Loss Plot of MobileNetV2



B) Confusion Matrix of MobileNetV2

Fig. 4 MobileNetV2 performance on maize disease classification

Classification Report:

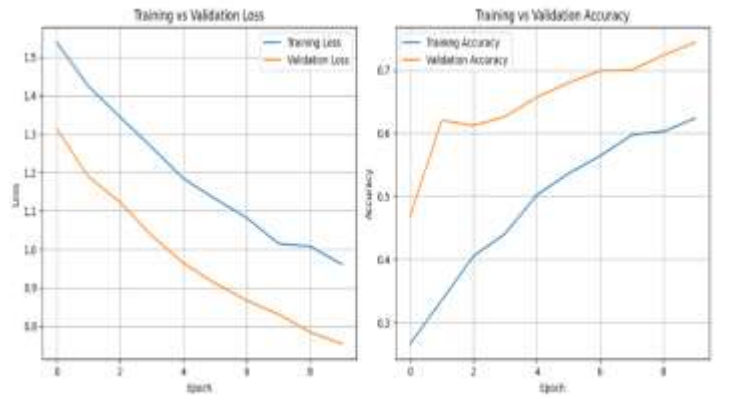
	precision	recall	f1-score	support
Blight	0.89	0.83	0.86	230
Common_Rust	0.98	0.95	0.96	262
Gray_Leaf_Spot	0.72	0.85	0.78	115
Healthy	0.99	1.00	1.00	233
accuracy			0.92	840
macro avg	0.89	0.91	0.90	840
weighted avg	0.92	0.92	0.92	840

Accuracy: 0.9166666666666666

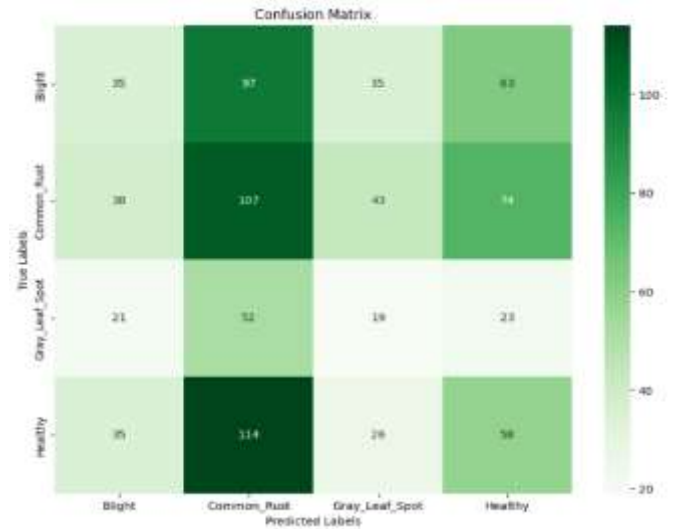
Table 3. Classification report of MobileNetV2

Fig. 5 shows how well VGG16 performed on the job of classifying maize leaf diseases. The model converges at a low accuracy level of about 26%, and the training and validation accuracy curves show a comparatively slow rising trend. Similarly, the loss curves show that the model does not achieve strong optimisation, indicating underfitting and inadequate feature learning, even though both training and validation loss reduce over epochs. This finding is corroborated by the categorisation report, which shows that VGG16 only achieved an overall accuracy of 26.1%. The model has the most difficulty correctly classifying maize leaf diseases among the classes. Gray_Leaf_Spot performed the worst, with a precision of 0.15 and recall of 0.17, whereas the Blight class achieved a precision of 0.27 and recall of 0.15.

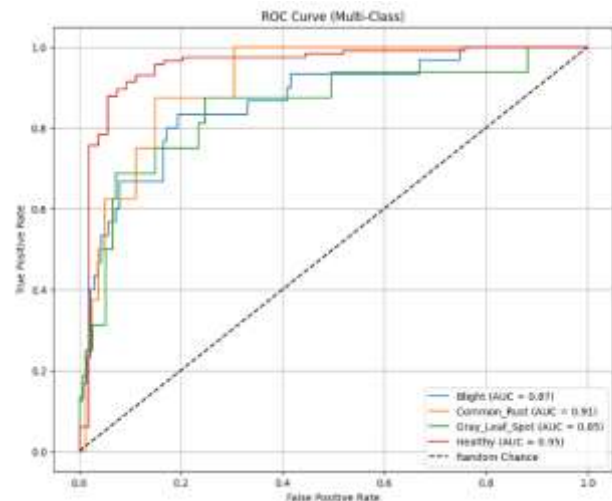
Despite being visually different, the Healthy class only managed to attain an accuracy of 0.27 and recall of 0.25, whilst Common_Rust performed marginally better with a precision of 0.29 and recall of 0.41. With comparatively low AUC scores across all categories, the ROC curve (Fig. 5) further supports the poor class separability. Numerous class misclassifications are also highlighted in the confusion matrix (Fig. 5). These findings show that, in comparison to alternative designs like MobileNetV2 and ResNet18, VGG16 performs poorly on the maize leaf disease dataset, exhibiting weak generalisation ability.



A) Accuracy and Loss Plot of VGG16



B) Confusion Matrix of VGG16



C) ROC curve of VGG16

Fig. 5 VGG16 performance on maize disease classification

Classification Report:

	precision	recall	f1-score	support
Blight	0.27	0.15	0.19	230
Common_Rust	0.29	0.41	0.34	262
Gray_Leaf_Spot	0.15	0.17	0.16	115
Healthy	0.27	0.25	0.26	233
accuracy			0.26	840
macro avg	0.25	0.24	0.24	840
weighted avg	0.26	0.26	0.25	840

Accuracy: 0.26071428571428573

Table 4. Classification report of VGG16

TEST PERFORMANCE

The models' final performance was assessed using an independent test set following training and validation. Images of maize leaves from all four categories—Blight, Common Rust, Grey Leaf Spot, and Healthy—that were not seen at all during training made up the test set. Before being fed into the trained model to produce predictions, each image was preprocessed using the same transformations used during validation. Evaluation measures like accuracy, precision, recall, and F1-score were calculated by comparing the anticipated and true labels. This testing phase is necessary to confirm the model's capacity for generalisation and to guarantee that the acquired characteristics can be used consistently in practical situations.

As illustrated in Fig. 6, the ResNet18 model had a test accuracy of 91.72% when assessed on the independent test set. This indicates that the model's generalisation extends considerably beyond the stages of training and validation. The confusion matrix showed that most samples were correctly classified, however because of their visual resemblance, Blight and Gray_Leaf_Spot were misclassified most often. Strong class discrimination was further supported by ROC analysis, which showed high AUC values in every category. Both accurate classifications and common errors are demonstrated by example forecasts, where Blight samples were sometimes incorrectly labelled as Common_Rust or Gray_Leaf_Spot. ResNet18's robustness as a contender for automated maize disease detection was confirmed by its consistent high accuracy identification of Healthy and Common Rust leaves in spite of these difficulties.

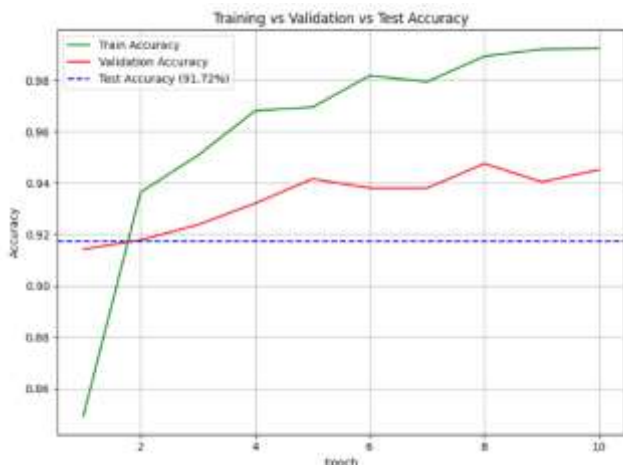


Fig. 6 Test accuracy of ResNet18

MobileNetV1 was evaluated using an independent test set of photos of maize leaves from the four classes (Blight, Common_Rust, Gray_Leaf_Spot, and Healthy). To ensure an objective assessment, the dataset was not used for either training or validation. After resizing and normalising each image, the trained network was used to classify it. The overall test accuracy was calculated by comparing the predictions with the true labels. This process evaluates how well the model reflects its performance in disease identification in the actual world and generalises to new data.

As illustrated in Fig. 7, the MobileNetV1 model had a test accuracy of 86.98%. Training and validation curves show that accuracy values stabilise after rapid convergence in the early epochs.

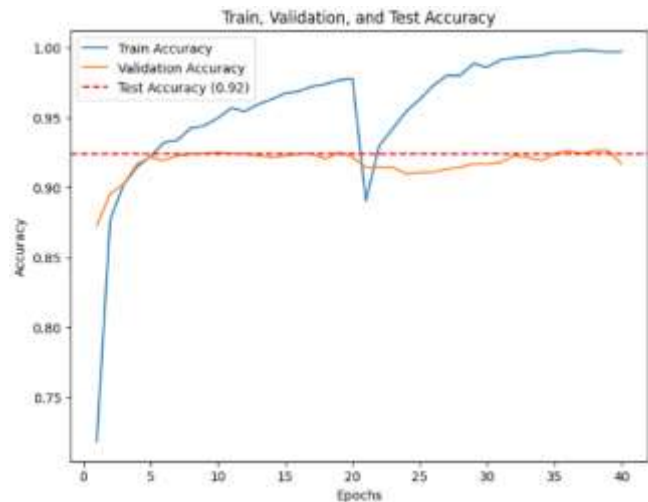


Fig. 7 Test accuracy of MobileNet16

An independent maize leaf test set comprising the four classes (Blight, Common_Rust, Gray_Leaf_Spot, and Healthy) was used to assess VGG16, much like the previous models. To ensure a fair assessment, none of these pictures were utilised for training or validation. After normalising and resizing each image to the 224 by 224 pixel VGG16 input size, the network was used to classify the images. To calculate evaluation measures, such as overall accuracy, classification report, and confusion matrix, the predicted labels were compared with ground truth labels. This process verifies VGG16's capacity to generalise to unknown field data.

As seen in Fig. 8, the VGG16 model obtained a 76% test accuracy rate. In contrast to ResNet18 and MobileNet, the training and validation accuracy curves show a slower rate of convergence with observable underfitting. VGG16's shortcomings in differentiating fine-grained leaf diseases were highlighted by the confusion matrix, which showed frequent misclassifications between Blight and Gray_Leaf_Spot. Although VGG16 performs worse than ResNet18 (91.72%) and MobileNet (86.98%), it still offers a respectable baseline accuracy percentage.



Fig. 8 Test accuracy of VGG16

Model	TA	VA	TstA
VGG 16	0.62	0.73	0.76
ResNet18	0.99	0.94	0.92
MobileNetV2	0.99	0.92	0.87

Table 5. Accuracy table of train, validate, and test

By the table 5. It is concluded that While its performance lags behind ResNet18 (91.72%) and MobileNet (86.98%), VGG16 still provides a reasonable baseline accuracy. However, due to its heavier architecture and lower accuracy, VGG16 is less suitable for real-world maize disease detection compared to more modern architectures optimized for efficiency and generalization.

V. APP DEVELOPMENT

The main objective of the suggested system is to employ artificial intelligence and contemporary frontend technologies to create a mobile application for the identification of maize (maize) leaf disease. This application's objective is to give farmers and other agricultural professionals a straightforward yet effective tool for detecting illnesses including common rust, blight, and grey leaf spot as well as verifying healthy conditions. Reducing crop losses and increasing production depend heavily on early disease identification, and this application uses image-based classification techniques to provide precise results instantly.

The app's architecture starts with picture acquisition, where the user can either upload a snapshot of the maize leaf from the gallery or use the device's camera to take one. The trained deep learning model then processes the chosen image and assigns the input to one of the predetermined categories. The processed data is shown in an interactive interface together with the

disease category, prediction confidence, and practical suggestions for preserving or enhancing crop health in the last phase, outcome visualisation. In Fig. 9, the system workflow is represented by a flowchart.

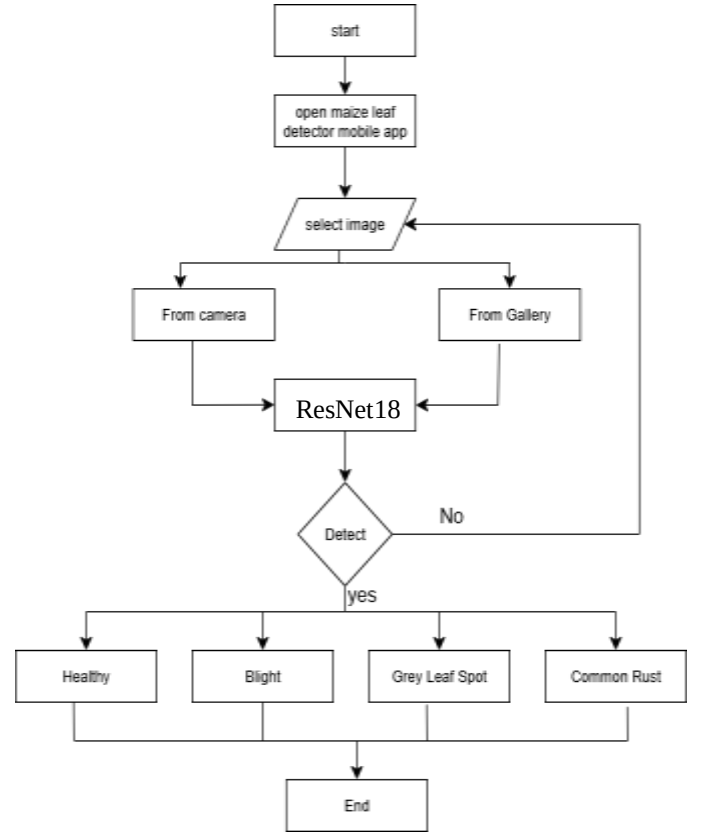


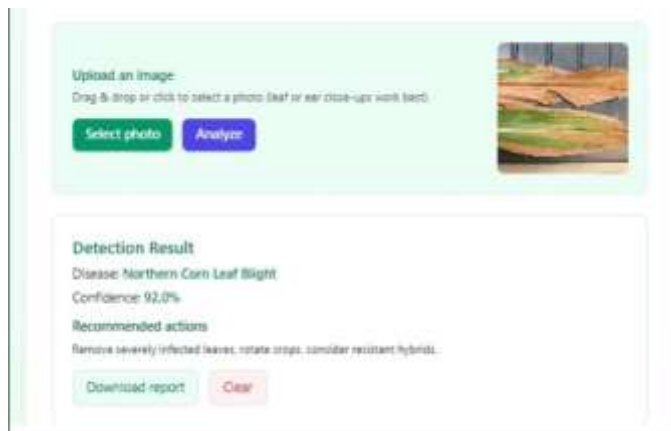
Fig.9 Flowchart of developing a final App

React, TypeScript, and Tailwind CSS are the main technologies used in this application. The interface is constructed using React, which offers modular and reusable UI elements using functional components and hooks like useState and useRef. Strong type checking is guaranteed by TypeScript, and shared interfaces provide uniformity during data transfers. Utility-based styling is made possible with Tailwind CSS, guaranteeing a tidy and mobile-friendly layout. Furthermore, the project is lightweight and effective due to the usage of Framer Motion for fluid interface animations and the contemporary ES module configuration with import maps, which eliminates the need for bundlers.

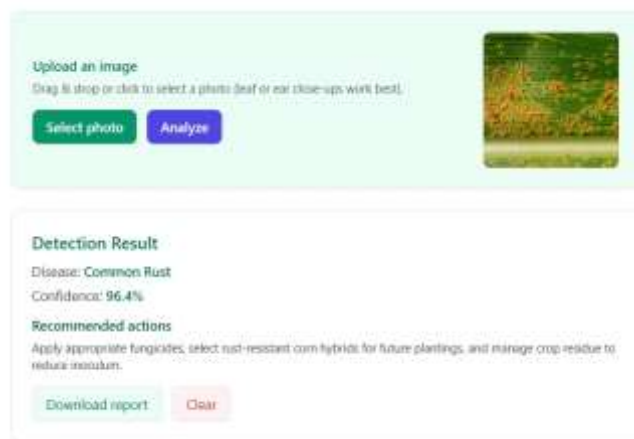
Components, services, and types make up the application structure. Diagnosis.tsx is used to upload and analyse photos, Hero.tsx is used for the landing page, and ResultCard.tsx is used to display the results. While the types module specifies the common data structures, the services layer controls communication between the frontend and the trained model. For user reference, the software also allows users to obtain comprehensive reports.

Results from experiments verify the app's efficacy. For instance, the system indicates that a maize leaf is healthy with 99.6% confidence when it is examined. It also suggests actions like keeping up excellent agricultural practices and keeping an eye on plant conditions as shown in fig. 10. The analysis dashboard is

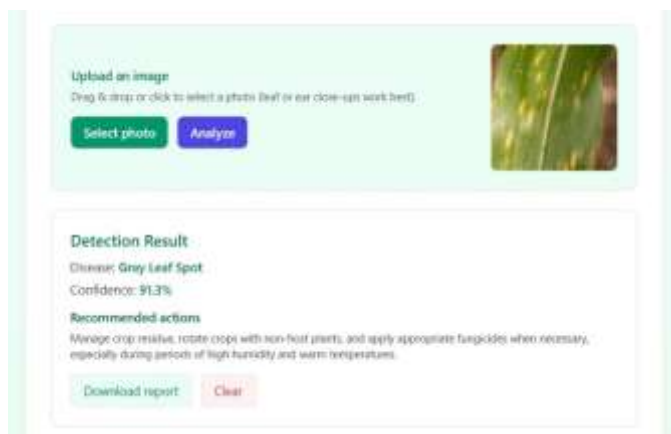
shown in another interface , where farmers can upload photographs fast and get a diagnosis right away.



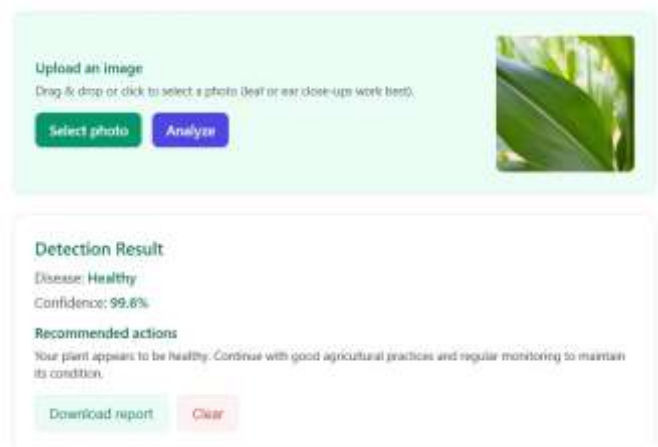
A)



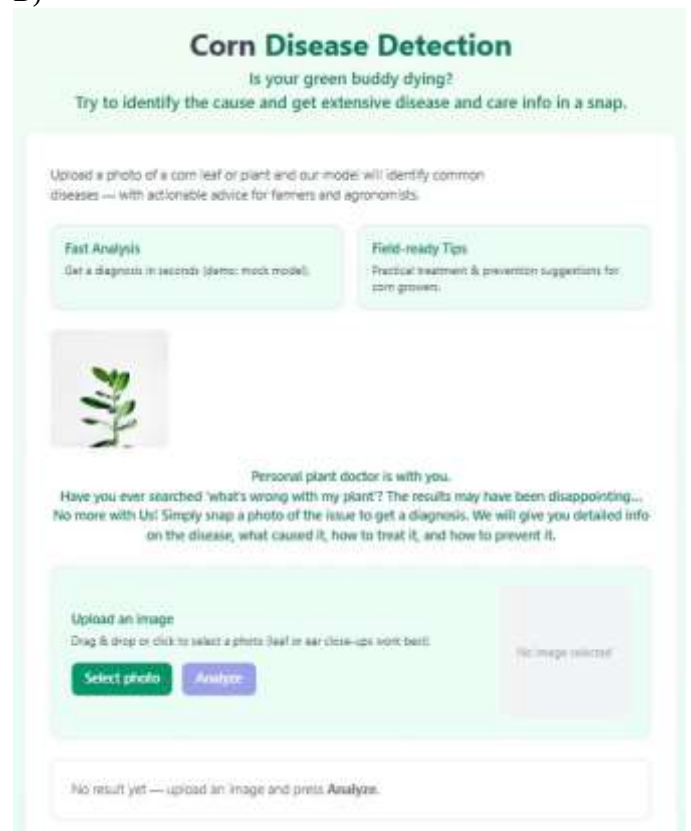
B)



C)



D)



E)

Fig. 10 A, B, C, D and E Sample maize leaf disease detection using the mobile app

In conclusion, this mobile application shows how artificial intelligence and contemporary web development technology may be used to solve important agricultural issues. The technology is easy to use, accurate, and lightweight, and it gives farmers useful information for efficiently managing maize diseases. This framework is a valuable and scalable option for smart agriculture since it can be expanded to accommodate numerous crops, offline usage, and interface with farm management systems.

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